Farbod Siahkali

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Ph.D. researcher in machine learning and control developing reliable perception and planning for autonomous systems. Research spans conformal prediction, multi-sensor fusion, multi-agent trajectory optimization, and safe reinforcement learning.

Education

Purdue University, Ph.D. in Electrical Engineering — Expected May 2029
Purdue University, M.S. in Electrical Engineering, GPA: 3.83 — Expected December 2026
University of Tehran, B.S. in Electrical Engineering (Control), GPA: 3.92 — July 2023

Publications & Manuscripts

- **Active Sensing and Deferred-Decision Trajectory Optimization for Robust Target Identification**. IEEE Control Systems Letters (L-CSS), 2026. CDC 2026 presentation accepted. First author.
- **Conformal Prediction-Based Multi-Sensor Classification**. Neurocomputing, vol. 678, article 133182, 2026. First author.
- **Multi-Agent Deferred-Decision Trajectory Optimization with Anytime-Valid Target Retention for Moving Candidates**. IEEE Robotics and Automation Letters (RA-L), under review. First author.
- **Coverage Guarantees for Pseudo-Calibrated Conformal Prediction under Distribution Shift**. IEEE Signal Processing Letters, under review, 2026. arXiv:2602.14913. First author.
- **Policy Gradient with Online Parameter Estimation and Evolving CBF Action Sets**. In preparation for submission to IEEE Control Systems Letters (L-CSS). First author.
- **Deep Reinforcement Learning for Epidemic Control of a Networked SIS Model**. IEEE Conference on Control Technology and Applications (CCTA), 2025.
- **Towards Opinion Shaping: A Deep Reinforcement Learning Approach in Bot-User Interactions**. arXiv:2409.11426, preprint, 2024. First author.
- **Image-Based and Partially Categorical Annotating Approach for Pedestrian Attribute Recognition**. SSRN, preprint, 2023.
- **SIVD: Dataset of Iranian Vehicles for Real-Time Multi-Camera Video Tracking and Recognition**. 8th Iranian Conference on Signal Processing and Intelligent Systems (ICSPIS), 2022. First author.

Research Experience

Graduate Research Assistant | Purdue University, Prof. Vijay Gupta — Aug 2024 – Present

- Reduced prediction-set size by **10–25%** at fixed coverage using conformal late fusion for dependent sensor outputs on synthetic, CIFAR-10, CIFAR-100, and MELD benchmarks.
- Developed slack-inflated and source-tuned pseudo-calibration algorithms for domain adaptation on MNIST and CIFAR benchmarks. Derived coverage bounds using Wasserstein distance under label-conditional covariate shift.
- Developed single- and multi-agent DDTO for target identification and interception using mixed-integer conic optimization, conformal prediction, sequential candidate rejection, and receding-horizon planning.
- Developed barrier-continuation policy gradient (BCPG) for safe reinforcement learning with online parameter estimation and control barrier functions (CBFs). Evaluated adaptive cruise control and planar navigation.

Research Assistant | University of Tehran, Intelligent Networks Lab — Oct 2023 – Jul 2024

- Implemented DDPG policies for networked epidemic control and budget-constrained opinion shaping.

Research Assistant | University of Tehran, Human & Robot Interaction Lab — May 2021 – Oct 2022

- Built PyTorch pipelines with YOLOv5 and OSNet for detection, re-identification, and multi-object tracking. Created CA-Duke labels for 32,000+ images and contributed to the 36,000+ image, 29-class SIVD dataset.

Technical Skills

Languages & Tools: Python, MATLAB, C, Git, Linux, LaTeX.
ML Libraries: PyTorch, TensorFlow, NumPy, Pandas, SciPy, scikit-learn.
Methods: Conformal prediction, uncertainty quantification, sensor fusion, Bayesian inference, domain adaptation, mixed-integer/conic optimization, trajectory planning, multi-agent systems, safe reinforcement learning, control barrier functions, computer vision.

Teaching, Service & Honors

Purdue ASPIRE REU research mentor (2026). University of Tehran teaching assistant (2020–2023). Reviewer: Neurocomputing, IEEE L-CSS, IEEE CDC. Best Undergraduate Project Award (2023).